

# DIVIT RAWAL

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**Research Interests.** Learning theory, high-dimensional and nonparametric statistics, structured learning, optimization.

## EDUCATION

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**University of California, Berkeley**

2023 – 2027

*B.A. Statistics, Computer Science, Physics*

*Berkeley, CA*

**Graduate-Level Coursework:** Statistical Learning Theory; Theoretical Statistics; Mathematical Statistics; Randomness and Computation; Learning for Dynamics and Control; Randomized Linear Algebra, Optimization, Large-Scale Learning.

**Teaching:** EE 122 (Communication Networks) Head TA (Sp26), Reader (Sp25). Developed lab curriculum, wrote and graded exams and homeworks, and led weekly recitation sections.

## PUBLICATIONS AND PREPRINTS

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[1] **Majority-of-Three is Optimal**

Divit Rawal, Nikita Zhivotovskiy.

*Under review, Transactions on Machine Learning Research (TMLR).* [arXiv]

[2] **Rao-Blackwellized Score Matching on Manifolds**

Divit Rawal.

*International Conference on Machine Learning (ICML) 2026, Workshop on Structured Probabilistic Inference and Generative Modeling.* [arXiv]

[3] **A Theory of Saddle Escape in Deep Nonlinear Networks**

Divit Rawal, Michael R. DeWeese.

*Under review, Journal of Machine Learning Research (JMLR).* [arXiv]

[4] **ALPHANSO: Open-Source Modeling of  $(\alpha, n)$  Neutron Source Terms**

Divit Rawal, Anthony J. Nelson, William Zywiec, Daniel Siefman.

*Nuclear Instruments and Methods in Physics Research, Section A.* [doi]

*Presented at ANS Student Conference 2026 and INMM Annual Meeting 2026.*

## AWARDS AND GRANTS

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**Best Paper Award, American Nuclear Society Student Conference**

Apr 2026

Best Paper in Mathematics, Computation, and AI Applications for “ALPHANSO: Open-Source Modeling of  $(\alpha, n)$  Source Terms.”

**VESSL AI Academia Grant**

Aug 2025

For independent research into geometric deep learning and information geometry; sole undergraduate recipient among cohort of Ph.D. students and postdoctoral researchers. (Supported work on [2].)

## RESEARCH EXPERIENCE

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**Berkeley Statistical Machine Learning Group**

Jun 2026 – Present

*Researcher* (advisor: Nikita Zhivotovskiy)

*Berkeley, CA*

Resolved conjecture establishing the majority vote of three consistent classifiers as an optimal PAC learner in the realizable setting; authored resulting paper [1].

Currently working on sample compression schemes and efficient optimal online learning algorithms.

**Berkeley Artificial Intelligence Research (BAIR)**

Sep 2024 – Present

*Researcher* (advisor: Michael DeWeese)*Berkeley, CA*

Empirically observed existence of in-context learning behavior in kernel ridge regression and analyzed emergence via spectral and random matrix theory methods.

Derived layer-imbalance identity for nonlinear networks, extending deep-linear network analysis and predicting first feature acquisition time in small-initialization neural networks; authored resulting paper [3].

**Lawrence Livermore National Laboratory**

Mar 2025 – Present

*Researcher, Nuclear Science and Security Consortium Fellow* (advisor: Daniel Siefman)*Livermore, CA*

Developed ALPHANSO [4], an open-source framework for deterministic modeling of  $(\alpha, n)$  neutron production using continuous slowing-down approximations and modern evaluated nuclear data.

Developing new capabilities for FREYA event-by-event fission model, improving simulation of correlated neutron and gamma emission for radiation transport applications.

**INVITED TALKS**

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**ALPHANSO: Open-Source  $(\alpha, n)$  Neutron Source Terms**

Lawrence Livermore National Laboratory

Mar 2026

Lawrence Berkeley National Laboratory

Apr 2026

University of Tennessee, Knoxville

Jun 2026

**INDUSTRY EXPERIENCE**

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**Cisco Systems, Foundation AI**

Jun – Aug 2025

*Software Engineer Intern**San Francisco, CA*

Developed components of the post-training and evaluation pipeline for cybersecurity-specialized LLMs based on Llama 3.1 8B, including instruction tuning and RLHF.

Contributed to the open-weight release of Foundation-Sec-8B and Foundation-Sec-8B-Instruct, including technical reports on data curation, training methodology, and benchmark evaluation.

**Amazon**

Aug – Dec 2023

*Software Engineer Intern**Remote*

Diagnosed and resolved production pipeline failures affecting 1M+ users; implemented a  $k$ -means clustering module in Java for OpenSearch ml-commons.

**TECHNICAL SKILLS**

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**Programming:** Proficient in Python, C, C++, Java, SQL; intermediate in MATLAB, FORTRAN, Julia, R.

**ML/AI:** Proficient in PyTorch; intermediate in JAX.

**Other:** Intermediate proficiency in Lean, Mathematica.